

Lilly Liu

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Intro

I am a software engineer who's led org-wide, broadly impactful AI initiatives at a fast-paced startup. I have built company-wide platform tools, designed autonomous agent workflows, and driven engineering velocity through scalable AI infrastructure.

Education

University of California, Berkeley | Double Major in Data Science and Economics

May 2024

- Activities: Drawn to Scale A Cappella (President), CalHacks, Association of Women in Engineering, Data Discovery Research Program

Fudan University | Study Abroad Program in Information and Computer Science

August 2023 - December 2023

Work Experience

Software Engineer, AI Native -- [Cambridge Mobile Telematics](#)

August 2024 – Present

- Developed an **autonomous Claude SDK agent harness** that replicated key stages of the SDLC by translating Jira tickets directly into multi-environment Pull Requests. Implemented AWS DynamoDB and cloud infrastructure for shared memory and an internal MCP layer for inter-agent communication. Deployed across 5+ teams to **boost PR velocity by 20%** with a 70% human approval rate on first pass.
- Delivered an **MCP orchestration platform** enabling self-serve AI agents and Slack chatbots across the org (including an agentic IRT triager, natural language-to-SQL agent, and general internal knowledge assistant) with built-in auth, MCP server registry, conversational memory, and self-learning.
- Ideated and architected an external MCP server for SDK documentation, streamlining SDK integration for enterprise clients, including DoorDash and Intact, reducing customer onboarding time from **days to hours**. As part of this effort, I designed and implemented the core authentication server for all production-facing external MCPs, leveraging AWS Cognito User Pools with OAuth 2.0 and PKCE.
- Spearheaded an org-wide "AI power user" initiative by architecting Sunfish, an internal engineering metrics platform (similar to Jellyfish); drove active AI tool adoption from **33% to 47%** in one quarter by deploying a central Claude Code marketplace, authoring core documentation, and leading bootcamps for senior engineers.
- Built and led adoption of a custom Python backend build system, unifying per-service custom scripts into a self-service command line tool integrated with DevOps and CI/CD pipelines. Reduced new service setup time from days to hours.

Software Engineering Intern -- [Brown Brothers Harriman & Co.](#)

June 2023 – September 2023

- Refactored a financial web application from JSP to Angular and Spring, improving page load time by 40%.

Software Engineering Intern -- [BAE Systems](#)

June 2022 – September 2022

- Built internal frontend features using TypeScript and REST APIs.
- Earned internal company award for organizing and leading the annual Bike Drive fundraiser.

Head Tutor and Course Liaison -- [UC Berkeley Student Learning Center](#)

January 2022 – December 2022

- Provided tutoring for hundreds of college students in Macroeconomics, assisted in the preparation and facilitation of exam reviews and tutoring material, utilized a variety of teaching methods to ensure student comprehension.

Academic Intern -- [UC Berkeley Division of Computing, Data Science, and Society](#)

January 2021 - December 2022

- Answered questions about lab/lecture content for UC Berkeley's Foundations of Data Science course (Data 8).

Projects

Build Your Own World | Java

- Created an engine for generating explorable worlds in 3000 lines, similar to the tile-based game Brogue.
- Used Disjoint Sets data structure in order to successfully connect rooms and hallways.

BearMaps | Java

- Combined data structures (KD-trees, Extrinsic MinPQ) and A* algorithm in order to implement a web-browser mapping application called BearMaps. Built an artificial intelligence that can solve arbitrary state space traversal problems using A* algorithm. Wrote original timing and logic tests.

Skills: Python, TypeScript, SQL, React, Model Context Protocol (MCP), LangGraph/LangChain, Claude Code, AWS, Cloud Infrastructure, DevOps, CI/CD, Docker, OAuth2/PKCE, Datadog, REST APIs, Git

Certifications: CodePath Advanced Software Engineering 2022, DeepLearning.AI Tensorflow Developer 2024

Interests: AI Native tools ▪ Full-stack Development ▪ Product Management ▪ Leadership ▪ Customer Solutions